Hydration Free Energy Calculations using the Drude Force Field with CHARMM

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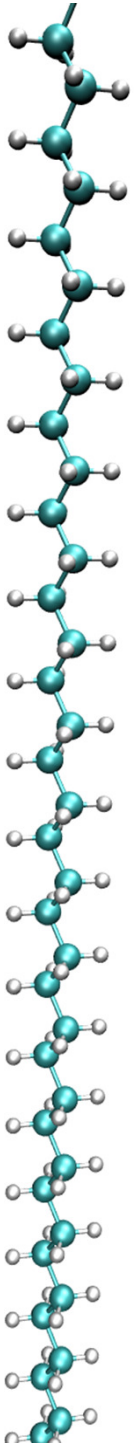


A New Model for NMA

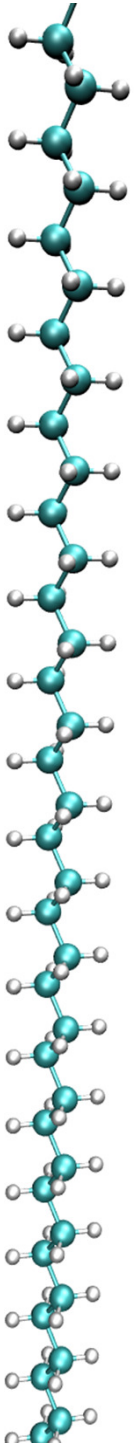
- The new NMA model was developed using a revised charge fitting scheme developed by Lei Huang
  - charges, polarizabilities, Thole parameters, and anisotrophies are optimized simultaneously
  - polarizabilities are restrained near Miller polarizabilities

Old

```
ATOM CL   CD33C   -0.300  ALPHA -2.511  THOLE 0.254
ATOM HL1  HDA3A    0.058
ATOM HL2  HDA3A    0.058
ATOM HL3  HDA3A    0.058
ATOM C    CD2O1A   0.682  ALPHA -1.576  THOLE 0.886
ATOM O    OD2C1A  -0.009  ALPHA -1.167  THOLE 1.310
ATOM N    ND2A2   -0.465  ALPHA -1.810  THOLE 1.149
ATOM H    HDP1A    0.325
ATOM CC   CD33C   -0.065  ALPHA -2.020  THOLE 0.254
ATOM HC1  HDA3A    0.094
ATOM HC2  HDA3A    0.094
ATOM HC3  HDA3A    0.094
ATOM LPA  LP      -0.3425
ATOM LPB  LP      -0.2815
ANISOTROPY O C LPA LPB  A11 0.88    A22 1.32
```

Lei's Code

```
ATOM CL   CD33C   -0.249  ALPHA -1.584  THOLE 0.330
ATOM HL1  HDA3A    0.050
ATOM HL2  HDA3A    0.050
ATOM HL3  HDA3A    0.050
ATOM C    CD2O1A   0.784  ALPHA -1.748  THOLE 1.517
ATOM O    OD2C1A   0.035  ALPHA -1.032  THOLE 0.915
ATOM N    ND2A2   -0.585  ALPHA -1.544  THOLE 0.438
ATOM H    HDP1A    0.297
ATOM CC   CD33C    0.119  ALPHA -1.229  THOLE 0.738
ATOM HC1  HDA3A    0.041
ATOM HC2  HDA3A    0.041
ATOM HC3  HDA3A    0.041
ATOM LPA  LP      -0.418
ATOM LPB  LP      -0.256
ANISOTROPY O C LPA LPB  A11  0.82322 A22  1.14332
```

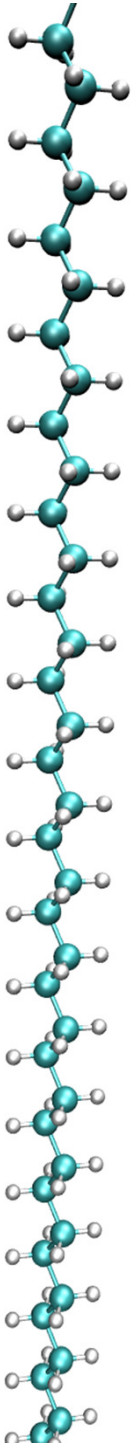


Lennard-Jones Terms

- Pedro found new Lennard-Jones parameters for these electrostatics
- Physical properties must be reevaluated with new model
  - hydration free energy

Atom Type		Emin (kcal/mol)	Rmin (Å)
N	Old	-0.260	1.925
	New	-0.200	1.800
O	Old	-0.180	1.900
	New	-0.200	1.750
C (sp^2)	Old	-0.100	1.600
	New	-0.050	1.775
C (methyl)	Old	-0.070	1.740
	New	-0.078	1.940

NBFIX		Emin (kcal/mol)	Rmin (Å)
N – ODW	Old	-0.2250	3.6569
	New	-0.2054	3.6569
	Comb.	-0.2341	3.712
O – ODW	Old	-0.2054	3.7369
	New	-0.2054	3.6269
	Comb.	-0.2202	3.5369

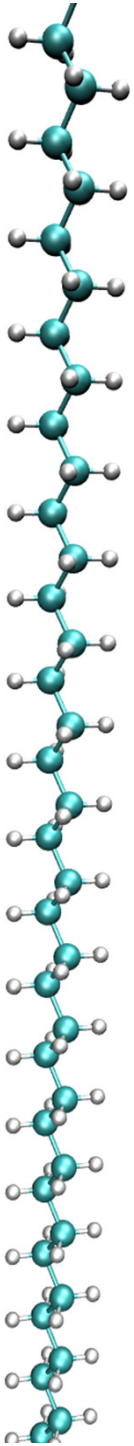


The Problem

- Procedures by the MacKerell and Roux groups gave hydration free energies that differed by ~1.5 kcal/mol for the hydration energy of NMA

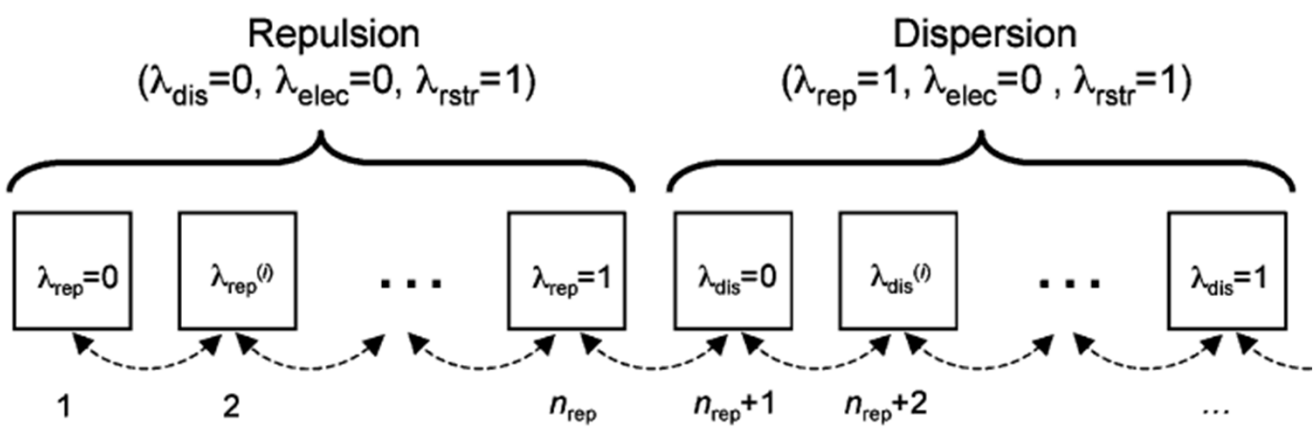
Ethanol	ΔG_{rep}	ΔG_{att}	ΔG_{chg}	ΔG_{solv}
Lopez	14.23	-13.00	-10.35	-9.11
Rowley	14.74	-13.11	-10.22	-8.59
Exptl				-10.1

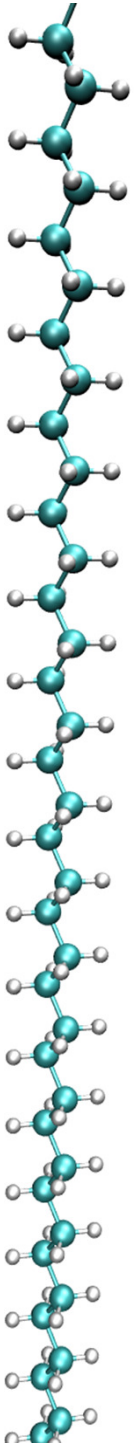
*LRC estimated at -0.45 kcal/mol



Hamiltonian Exchange Replica Exchange MD

- 18 Repulsive Energy Windows
  - scaling intervals:
    - [0.0, 0.2, 0.3, 0.4, 0.5, 0.6, 0.7, 0.8, 0.9, 1.0]
  - λ=[0,1]
- 11 Attractive Energy Windows
- Charging component of Drude systems cannot be included in exchange
- Exchanges attempted every 100 fs
- ~50% acceptance rate



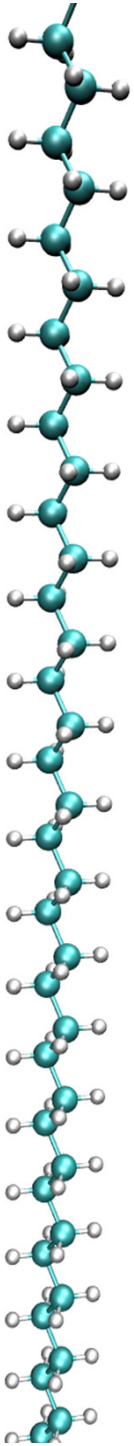


Thermostats Used in Gas Phase Component

- Monomer FEP was calculated using leap-frog Langevin integrator (NOT Dual thermostat)
- Results in ~0.1 kcal/mol errors
- There is now a functional dual-Langevin thermostat in CHARMM that works with VV2

```
TPCONTROL NTHER 2 NHGAM 5.0 NHGAMD 10.0 -
      THER 1 TREF 298.15 LANG SELECT .NOT. TYPE D* END -
      THER 2 TREF 1.0 LANG SELECT TYPE D* END

DYNAMICS  vv2       start    nstep @nstep  timestp @timestep  -
          ntrfrq @nstep  iprfrq  -1  -
          nprint    1000 iseed @useed -
          iasvel       1  firstt   @temp    finalt    @temp    -
          inbfrq      -1  imgfrq   -1     ihbfrq   0  ilbfrq    0  -
          iunread     -1  -
          iunwrite    -1  -
          iuncrd      -1  nsavcrd     100   -
          punit 52 wham 54
```

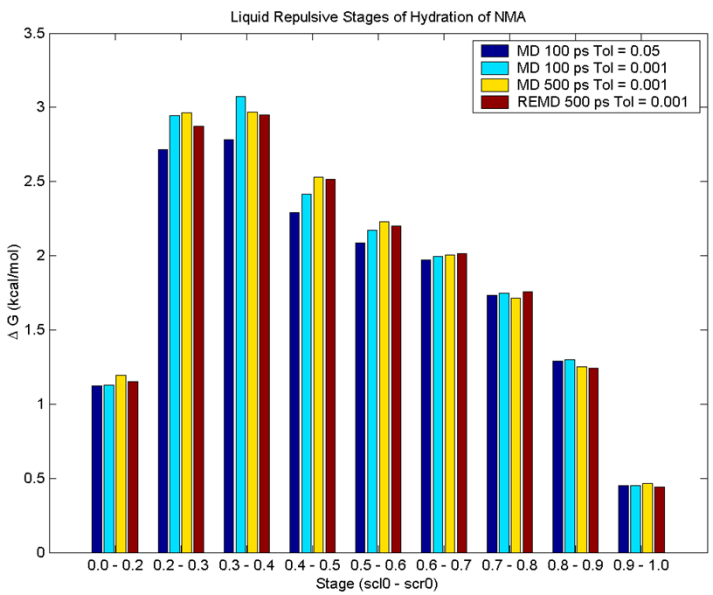


Integration of Charging Free Energy Component

- The charging free energies component
- The non-polar components can be integrated using WHAM
- Integrating the charging component with WHAM introduces an error ~0.6 kcal/mol

Run	Integration	ΔG (kcal/mol)
1	WHAM	-62.39
1	TI	-63.11
2	WHAM	-62.48
2	TI	-63.18

Difference in Repulsive Component is Due to WHAM SCF Tolerance



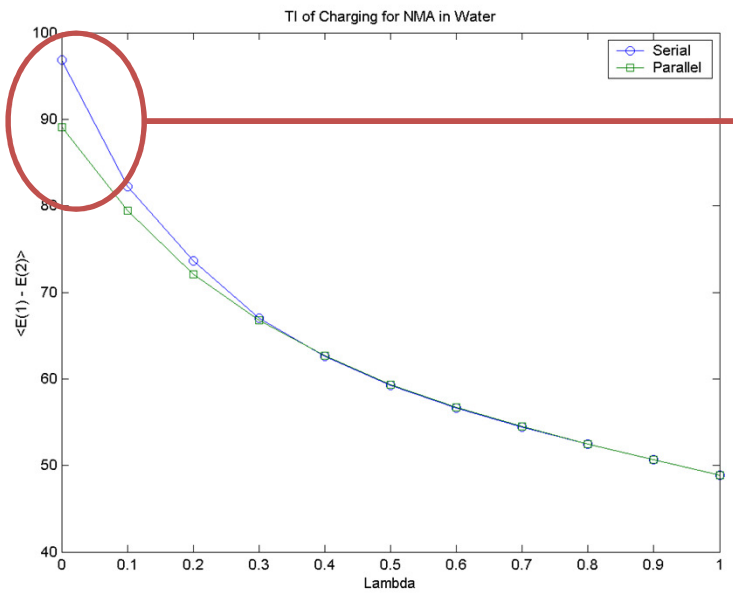
Simulation	WHAM Tolerance	ΔG^{rep} (kcal/mol)
MD 100 ps	0.050	16.45
MD 100 ps	0.001	17.22
MD 500 ps	0.001	17.32
REMD 500 ps	0.001	17.15

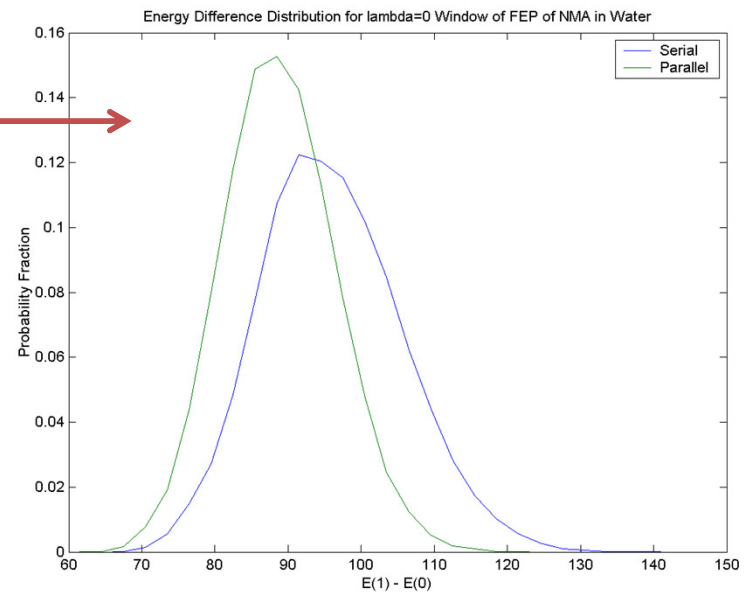
```
open read card unit 54 name @job.wham
WHAM MAXWindow 50 MAXTime @MTIME WSIZe @wsize  -
   unit 54 tol 0.05 nstep 200
close unit 54
write title unit 55
* job @job
* Wham free energy: ?WHAMFE
*
RETURN
```

```
open read card unit 54 name @job.wham
WHAM MAXWindow 50 MAXTime @MTIME WSIZe @wsize  -
   unit 54 tol 0.001 nstep 200
close unit 54
write title unit 55
* job @job
* Wham free energy: ?WHAMFE
```

Serial vs Parallel CHARMM

A small but significant difference between the parallel and serial versions of CHARMM was found in the charging free energy of NMA.

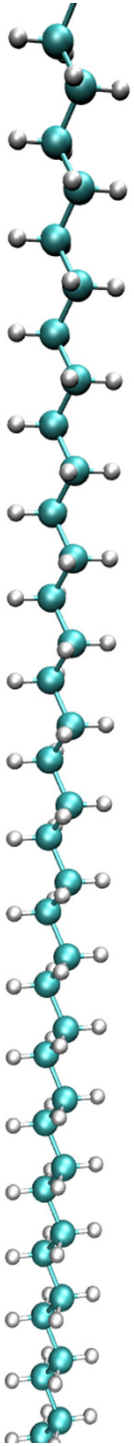




Serial vs Parallel Resolution

- A bug was present in `epert.src` that called the THOLE_NBX subroutine only once rather than once per process
- Serial and parallel CHARMM now give statistically equivalent results

```
IF (MYNOD == 0) THEN      !##PARALLELx
       IF (QDRUDE .AND. QETPRT(ELEC)) THEN
            CALL THOLE_NBX(ISDRUDE,ALPHADP,THOLEI, &
               PPCG, &
               ETPRTM(ELEC),BNBNDR%INB14,BNBNDR%IBLO14, &
               X,Y,Z,DX,DY,DZ,DD1,IUPT,QSECDL,NATOMP)
```

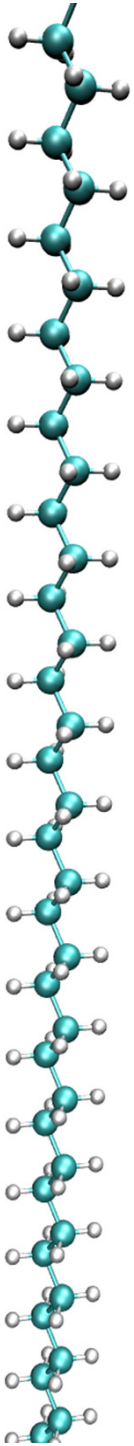


Results

- With the revised procedure the calculated hydration free energy of NMA is:

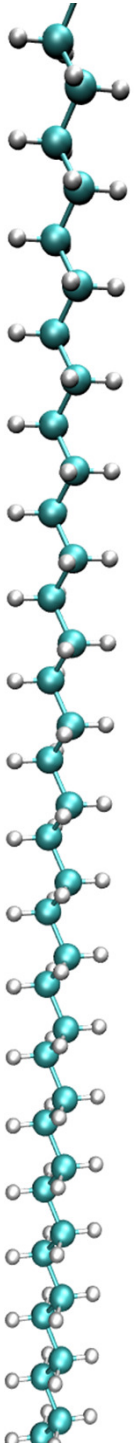
NMA (with NBFIX)	ΔG_{rep}	ΔG_{att}	ΔG_{chg}	ΔG_{solv}
Revised	14.92	-13.03	-10.98	-9.09
REMD	14.76	-13.07	-10.98	-9.30
Exptl				-10.1

NMA (without NBFIX)	ΔG_{rep}	ΔG_{att}	ΔG_{chg}	ΔG_{solv}
Revised	14.62	-12.69	-13.47	-11.54
REMD	14.65	-12.78	-13.47	-11.60
Exptl				-10.1



Part II: Revisiting Old Molecules

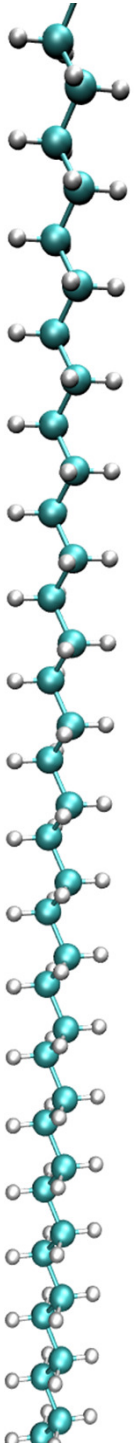
- I have used this procedure to recalculate the hydration free energies for some existing molecules in the Drude force field
- Test cases:
  - Benzene
  - Ethanol
  - Butane



Hydration Free Energy of Benzene

Benzene	ΔG_{rep}	ΔG_{att}	ΔG_{chg}	ΔG_{solv}
Published[1]	14.61	-14.24	-1.43	-1.06
Revised	15.47	-12.34	-2.53	0.60
REMD	15.18	-12.43	-2.53	0.22
Exptl				-0.86

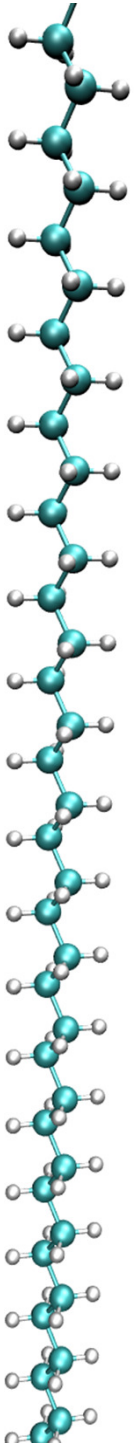
[1]Lopez et al. *J. Phys. Chem. B* 2007, 111, 2873-2885



Hydration Free Energy of Ethanol

Ethanol	ΔG_{rep}	ΔG_{att}	ΔG_{chg}	ΔG_{solv}
Published[1]				-5.66
Published (PS LJ)[2]				-5.38
Revised	11.61	-8.44	-7.11	-3.93
REMD	11.30	-8.46	-7.11	-4.26
Exptl				-5.01

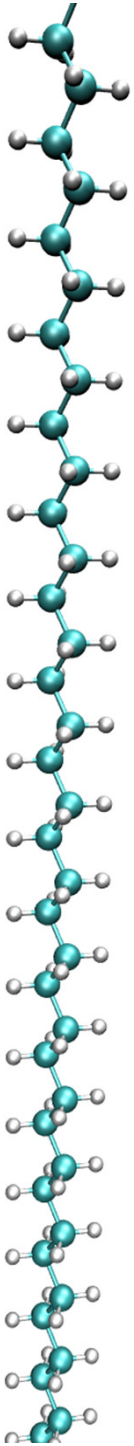
[1]Anisimov et al, *J. Chem. Theory Comput.* 2007, 3, 1927–1946
[2]Baker et al. *J. Chem. Theory Comput.* 2010, 6, 1181–1198



Hydration Free Energy of Butane

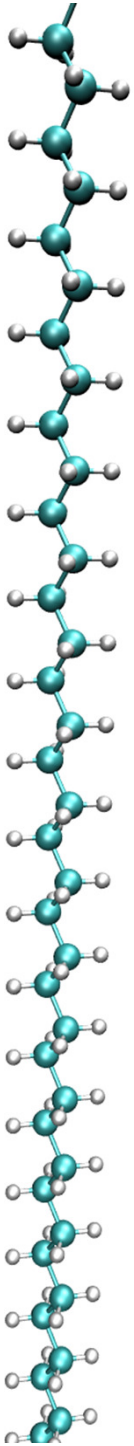
Ethanol	ΔG_{rep}	ΔG_{att}	ΔG_{chg}	ΔG_{solv}
Published[1]	14.12	-12.34	-0.31	1.46
Published (PS LJ)[2]				2.54
Revised	14.99	-11.16	-0.41	3.42
REMD	15.03	-11.21	-0.41	3.42
Exptl				2.15

[1]Vorobyov et al. *J. Phys. Chem. B* 2005, **109**, 18988–18999
[2]Baker et al. *J. Chem. Theory Comput.* 2010, 6, 1181–1198



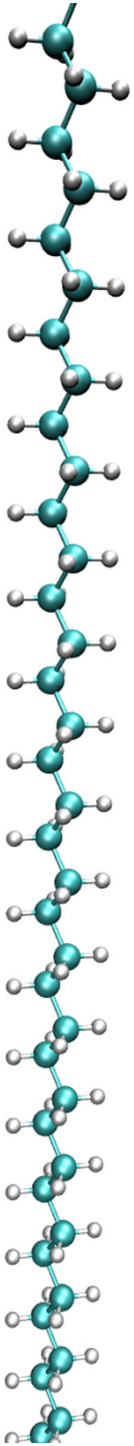
Charge Fitting of Benzene

Old	Lei's Code
ATOM CG CD2R6A -0.1106 ALPHA -1.615 THOLE 1.195	ATOM CG CA -0.085 ALPHA -1.820 THOLE 1.279
ATOM HG HDR6A 0.1106	ATOM HG HA 0.085
ATOM CD1 CD2R6A -0.1106 ALPHA -1.615 THOLE 1.195	ATOM CD1 CA -0.085 ALPHA -1.820 THOLE 1.279
ATOM HD1 HDR6A 0.1106	ATOM HD1 HA 0.085
ATOM CD2 CD2R6A -0.1106 ALPHA -1.615 THOLE 1.195	ATOM CD2 CA -0.085 ALPHA -1.820 THOLE 1.279
ATOM HD2 HDR6A 0.1106	ATOM HD2 HA 0.085
ATOM CE1 CD2R6A -0.1106 ALPHA -1.615 THOLE 1.195	ATOM CE1 CA -0.085 ALPHA -1.820 THOLE 1.279
ATOM HE1 HDR6A 0.1106	ATOM HE1 HA 0.085
ATOM CE2 CD2R6A -0.1106 ALPHA -1.615 THOLE 1.195	ATOM CE2 CA -0.085 ALPHA -1.820 THOLE 1.279
ATOM HE2 HDR6A 0.1106	ATOM HE2 HA 0.085
ATOM CZ CD2R6A -0.1106 ALPHA -1.615 THOLE 1.195	ATOM CZ CA -0.085 ALPHA -1.820 THOLE 1.279
ATOM HZ HDR6A 0.1106	ATOM HZ HA 0.085



Charge Fitting of Ethanol

Old	Lei's Code
ATOM C1 CD32A -0.060 **ALPHA -1.000** THOLE 1.3	ATOM C1 C3 0.138 **ALPHA -1.902** THOLE 1.134
ATOM O1 OD31A 0.000 ALPHA -1.028 THOLE 1.3	ATOM O1 OH 0.140 ALPHA -0.969 THOLE 1.140
ATOM HO1 HDP1A 0.360	ATOM HO1 HO 0.351
ATOM H1A HDA2A 0.080	ATOM H1A H1 0.010
ATOM H1B HDA2A 0.080	ATOM H1B H1 0.010
ATOM C2 CD33A -0.180 ALPHA **-1.400** THOLE 1.3	ATOM C2 C3 -0.047 ALPHA **-2.049** THOLE 1.142
ATOM H2A HDA3A 0.060	ATOM H2A HC 0.010
ATOM H2B HDA3A 0.060	ATOM H2B HC 0.010
ATOM H2C HDA3A 0.060	ATOM H2C HC 0.010
ATOM LP1A LP -0.230	ATOM LPAO1 LP -0.316
ATOM LP1B LP -0.230	ATOM LPBO1 LP -0.316
ANISOTROPY O1 C1 LP1A LP1B A11 0.8108 A22 1.2162	ANISOTROPY O1 C1 LPAO1 LPBO1 A11 0.70099 A22 1.119



Charge Fitting of Butane

Old	Lei's Code
Atom H11 HDA3A 0.059	ATOM H11 HC 0.049
Atom H12 HDA3A 0.059	ATOM H12 HC 0.047
Atom H13 HDA3A 0.059	ATOM H13 HC 0.047
Atom C1 CD33A -0.177 ALPHA -2.051 THOLE 1.3	ATOM C1 C3 -0.215 ALPHA -2.070 THOLE 1.028
Atom H21 HDA2A 0.078	ATOM H21 HC 0.010
Atom H22 HDA2A 0.078	ATOM H22 HC 0.010
Atom C2 CD32A **-0.156** ALPHA -1.660 THOLE 1.3	ATOM C2 C3 **0.053** ALPHA -1.943 THOLE 1.017
Atom H31 HDA2A 0.078	ATOM H31 HC 0.010
Atom H32 HDA2A 0.078	ATOM H32 HC 0.010
atom C3 CD32A **-0.156** ALPHA -1.660 THOLE 1.3	ATOM C3 C3 **0.053** ALPHA -1.943 THOLE 1.017
atom H41 HDA3A 0.059	ATOM H41 HC 0.047
atom H42 HDA3A 0.059	ATOM H42 HC 0.047
atom H43 HDA3A 0.059	ATOM H43 HC 0.047
atom C4 CD33A -0.177 ALPHA -2.051 THOLE 1.3	ATOM C4 C3 -0.215 ALPHA -2.070 THOLE 1.028